Local Plug-and-Play Compatible Stability Conditions for AC Grids

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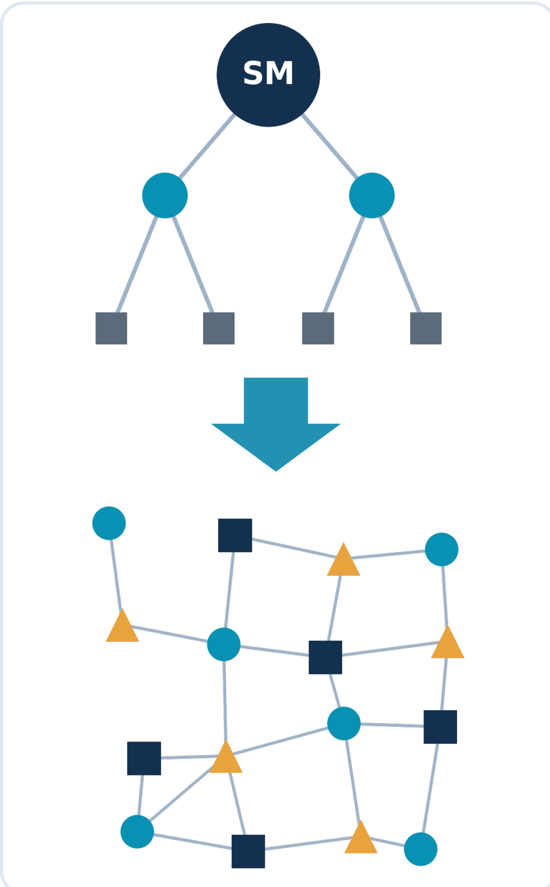
1 Motivation: the changing grid

- **Inertia is disappearing:** synchronous machines → grid-forming power electronics [1]
- New, **complex controls** acting on much faster timescales
- Topology in constant flux — **distributed, meshed, ever-changing**
- Traditional monolithic eigenvalue analysis no longer **scalable**

— REQUIREMENTS

Stability guarantees that do not require re-analysing the whole grid any time the network interconnection changes



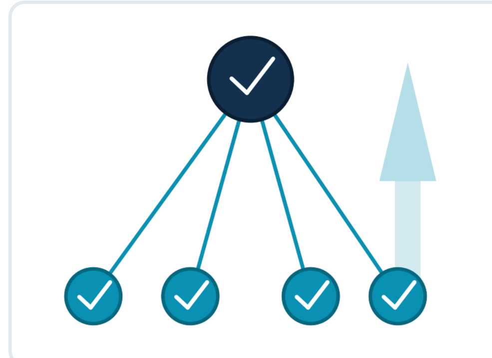
2 Why local conditions?

— AIM

Certify grid-wide small-signal stability via frequency-domain quadratic constraints on the input–output properties of local grid subsystems.

— BENEFITS

- Top-down study → **bottom-up**
- Basis for interconnection standards & **grid codes** [5]
- Frequency-domain ⇒ **black-box** inverter models
- **Plug-and-play:** a compliant device connects without global re-analysis

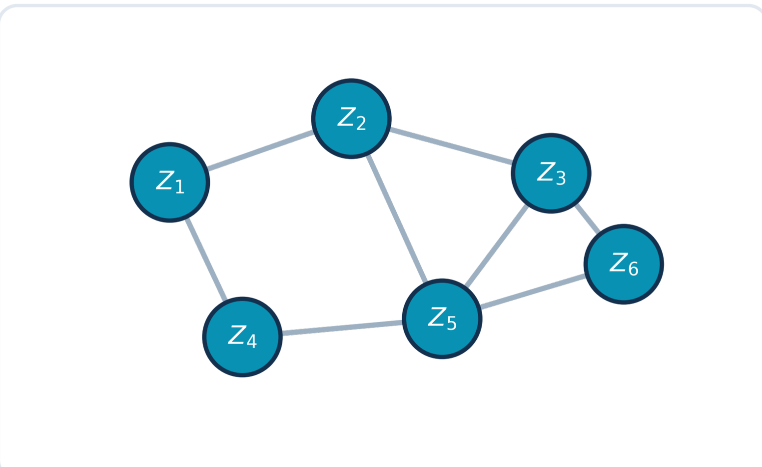


— OUR APPROACH

- ✓ Full network and bus dynamics (vs. [2], [3])
- ✓ Plug-and-play compatible (vs. [4])
- ✓ Mix conditions over frequencies and system formulations
- ✓ Graphical interpretations

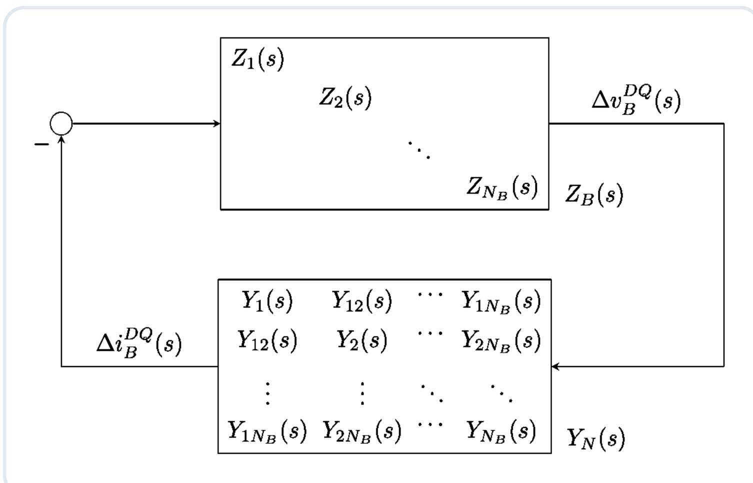
3 Grid model

— NETWORK GRAPH



- Graph: buses (nodes), lines (edges)
- Common **DQ** frame mapping AC sinusoids to constants at equilibrium

— FEEDBACK STRUCTURE



- Linearise ⇒ 2×2 impedance $Z_i(s)$
- Network: sparse admittance $Y_N(s)$
- Kirchhoff's laws ⇒ feedback pair

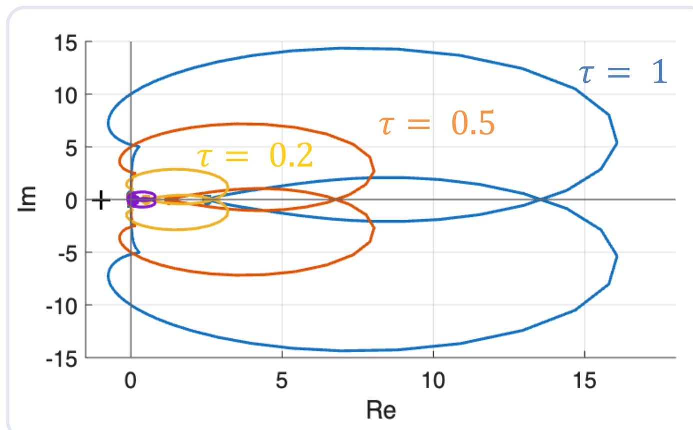
4 Nyquist criterion → quadratic constraints

- Generalised Nyquist criterion
- $Z_B(s)$ and $Y_N(s)$ stable ⇒ no encirclements

— HOMOTOPY

Sufficient condition:

$$-1 \notin \sigma(\tau L(j\omega)) \qquad \forall \tau \in [0,1], \;\; \forall \omega \geq 0$$



— QUADRATIC CONSTRAINTS

Holds at ω if multipliers $\Pi(j\omega)$ at **each** frequency satisfy two QMIs:

$$\begin{bmatrix} I \\ -Y_N(j\omega) \end{bmatrix}^* \begin{bmatrix} \Pi_{11} & \Pi_{12}^* \\ \Pi_{12} & \Pi_{22} \end{bmatrix} \begin{bmatrix} I \\ -Y_N(j\omega) \end{bmatrix} \leq 0 \qquad \begin{bmatrix} Z_B(j\omega) \\ I \end{bmatrix}^* \begin{bmatrix} \Pi_{11} & \Pi_{12}^* \\ \Pi_{12} & \Pi_{22} \end{bmatrix} \begin{bmatrix} Z_B(j\omega) \\ I \end{bmatrix} > 0$$

$$\Pi_{11} = \Pi_{11}^* \leq 0, \qquad \Pi_{22} = \Pi_{22}^* \geq 0$$

5 Decentralised conditions (bus level)

— EXPLOIT BLOCK DIAGONAL BUSES

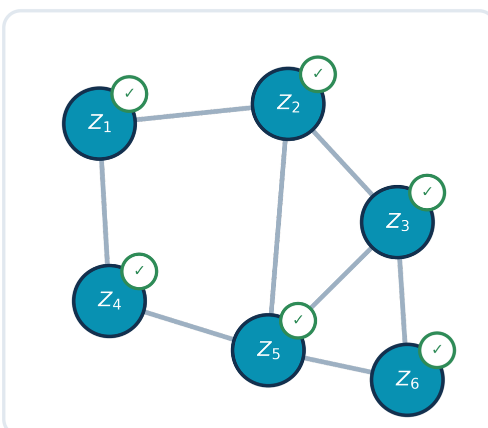
- Block-diagonal multipliers

  ⇒ bus QMI **decouples** into one condition per bus:

$$\begin{bmatrix} Z_i(j\omega) \\ I_2 \end{bmatrix}^* \begin{bmatrix} \Pi_{11}^i & \Pi_{12}^{i*} \\ \Pi_{12}^i & \Pi_{22}^i \end{bmatrix} \begin{bmatrix} Z_i(j\omega) \\ I_2 \end{bmatrix} > 0 \qquad \forall \nu_i \in \mathcal{V}_B$$

- Blocks independent of non-local parameters

  ⇒ **fully decentralised, plug-and-play**.



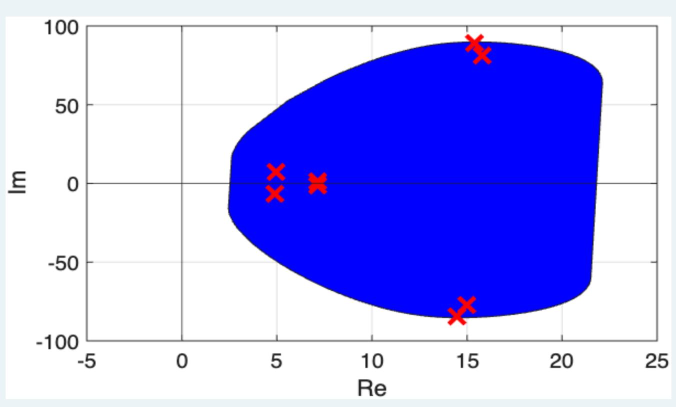
— POSITIVE REAL

$Y_N(s)$ positive-real [3]

⇒ $\Pi_{11} = \Pi_{22} = 0, \Pi_{12} = I_2$

$$Z_i(j\omega) + Z_i(j\omega)^* > 0$$

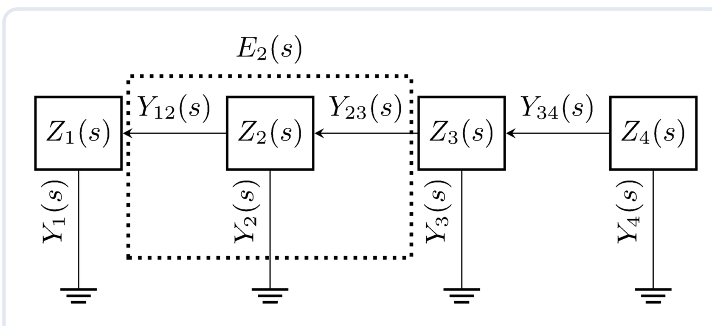
Graphical interpretation: numerical range of $Z_i(j\omega)$ lies in the open right half-plane

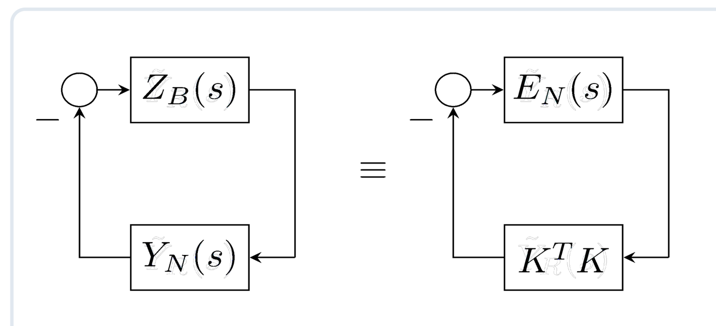


6 Extended subsystem conditions

— EXPLOIT SPARSE NETWORK

- Regroup each bus with its lines/loads into an **extended subsystem**
- Feedback loop with static interconnection $K^T K$
- Loci-equivalent ⇒ test **either** loop.





— CONSTRAINTS ON SUBSYSTEMS

Per-line constraint

$$-\Pi_{12}^{ij} - \Pi_{12}^{ij\,*} + 2\,\Pi_{22}^{ij} \leq 0$$

Per-bus constraint, $\forall \nu_i \in \mathcal{V}_B$

$$\begin{bmatrix} E_i(j\omega) \\ I \end{bmatrix}^* \begin{bmatrix} 0 & \hat{\Pi}_{12}^{i\,*} \\ \hat{\Pi}_{12}^i & \hat{\Pi}_{22}^i \end{bmatrix} \begin{bmatrix} E_i(j\omega) \\ I \end{bmatrix} > 0$$

Locality preserved: network changes impact nearby systems ⇒ plug-and-play.

7 Main result & case study

— PROPOSITION

The grid is stable if, at every $\omega \geq 0$,

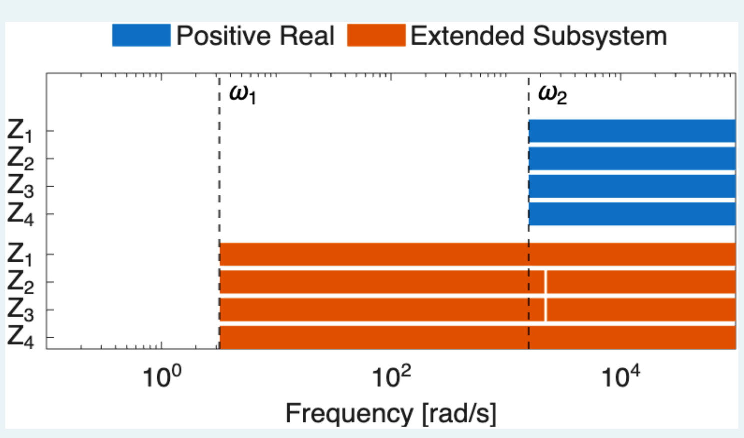
- decentralised bus conditions; OR
- extended subsystem conditions

hold, and the choice may differ across frequency ranges.

— IMPLICATIONS

- Basis for plug-and-play grid code
- Mix conditions and formulations across frequencies

— CASE STUDY: KUNDUR [6]



- Decentralised conditions across problematic frequencies
- Lower frequencies: trivially satisfied

8 Conclusions & future work

— CONTRIBUTIONS

- ✓ Scalable, fully **local** certification of small-signal stability via quadratic constraints
- ✓ Foundation for **plug-and-play** compatible grid codes
- ✓ **Reduced conservatism** via mixed conditions and multiple grid formulations
- ✓ Unsimplified bus and network dynamics
- ✓ Graphical interpretations

— EXTENSIONS

- Accounting for unstable bus systems
- Power-flow representation at low frequencies to enable Nyquist analysis

— REFERENCES

[1] F. Milano et al., "Foundations and Challenges of Low-Inertia Systems," PSCC, 2018.

[2] V. Häberle et al., "Decentralized Parametric Stability Certificates for Grid-Forming Converter Control," arXiv:2503.05403, 2025.

[3] J. D. Watson, Y. Ojo, K. Laib, I. Lestas, "A Scalable Control Design for Grid-Forming Inverters in Microgrids," IEEE Trans. Smart Grid, 2021.

[4] L. Huang et al., "Gain and Phase: Decentralized Stability Conditions for Power Electronics-Dominated Power Systems," IEEE Trans. Power Syst., 2024.

[5] D. Kong, A. Johnson, X. Zhou, "Best Practice: Grid-Forming Converter Technological Development in Great Britain," IEEE Power & Energy Mag., 2025.

[6] P. S. Kundur, O. P. Malik, Power System Stability and Control, 2nd ed., McGraw-Hill, 2022.